

Generative AI Internship — AI Workflow Assignment

1. Objective

The objective of this assignment is to document practical **Generative AI workflows** and understand how different AI concepts work together to create useful applications.

The workflows focus on:

- User Input
 - Prompt Engineering
 - Large Language Models (LLMs)
 - LLM APIs
 - Structured Outputs
 - Final AI Response
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2. Concepts Practiced

- Generative AI
 - Large Language Models (LLMs)
 - Tokens
 - Context Windows
 - Prompt Engineering
 - LLM APIs
 - Structured Outputs
 - AI Workflow Design
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3. AI Workflow Examples

Example 1 — AI Question-Answering Workflow

Workflow:

User Input → Prompt → LLM → Generated Answer

Description:

The user submits a question. The application places the question into an appropriate prompt and sends it to an LLM through an API. The LLM processes the input and generates an answer that is returned to the user.

Key Learning:

I learned how a basic AI application connects user input with an LLM and uses the generated response as the final application output.

Example 2 — AI Text Analysis Workflow

Workflow:

Text Input → Prompt Engineering → LLM → Structured Output → Analysis

Description:

The user provides text that needs to be analyzed. A carefully designed prompt tells the LLM what information to extract or analyze. The LLM processes the text and returns the result in a structured format.

For example, the workflow could extract:

- Key points
- Categories
- Sentiment
- Important information
- Specific fields from the text

Key Learning:

I learned that **structured outputs** make AI responses more predictable and easier to process inside software applications.

Example 3 — AI Content Generation Workflow

Workflow:

User Requirements → Prompt → LLM → Structured Response → Final Content

Description:

The user provides requirements such as:

- Topic
- Target audience
- Tone
- Content format
- Required information

These requirements are converted into a clear prompt. The LLM generates content based on the instructions, and the application can format the response into the final output.

Key Learning:

I learned that **prompt design has a direct impact on the quality, relevance, and structure of an LLM's response.**

4. General AI Workflow

A general Generative AI workflow can be represented as:

User Input → Prompt Engineering → LLM/API → Structured Output → Application Response

Each stage has a specific purpose:

1. **User Input** — Provides the information or request.
 2. **Prompt Engineering** — Converts the request into clear instructions.
 3. **LLM/API** — Processes the prompt using a Large Language Model.
 4. **Structured Output** — Organizes the generated information into a predictable format.
 5. **Application Response** — Presents the result to the user.
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5. What I Learned

One important lesson from these experiments is that building an AI application is not only about calling an LLM.

The complete workflow is also important. **Clear inputs, well-designed prompts, appropriate context, and structured outputs** can make AI functionality more predictable and easier to integrate into real software applications.

I also learned that different workflows can use the same LLM technology for different purposes, such as answering questions, analyzing information, or generating content.

6. Technologies and Concepts

Area	Concept
Artificial Intelligence	Generative AI
Models	Large Language Models
API	LLM APIs
Prompting	Prompt Engineering
Output	Structured Outputs
Architecture	AI Workflows
Context	Tokens and Context Windows

7. Next Learning Goals

After completing these workflow experiments, my next learning goals are:

- AI Agents
 - Tool Calling
 - Multi-step AI Workflows
 - Agent Orchestration
 - Multi-Agent Systems
 - Real-world AI-powered applications
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8. Internship Context

This assignment is part of the **Generative AI Internship at Big Brains**.

The purpose of this activity is to document practical learning and demonstrate how Generative AI concepts can be combined into workflows for real-world software applications.

9. Evidence of Work

Add **1–2 screenshots** of the actual work completed during the internship.

Screenshot 1: LLM/API or Prompt Engineering experiment

Screenshot 2: Structured Output or AI Workflow experiment

Make sure screenshots do not contain API keys, passwords, private information, or other sensitive data.